

JOHNATHAN AHDOUT

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EDUCATION

Princeton University

B.A. Mathematics; Minors: Computer Science & Optimisation and Quantitative Decision Science

Princeton, NJ
Present – May 2029

- GPA: 3.79/4.00

- Relevant Coursework: *Honours Linear Algebra, Single-Variable Real Analysis, Probability Theory, Data Structures and Algorithms, Abstract Algebra, Combinatorics, Financial Investments & Derivatives, Multivariable Calculus, Fundamentals of Statistics*

John L. Miller Great Neck North High School

High School Diploma

Great Neck, NY
Sep 2021 – Jun 2025

- Salutatorian; GPA: 97.8/100; SAT: 1590 (800 Math, 790 Reading & Writing)

RESEARCH & EXPERIENCE

Dickson Lab @ Michigan State University, ACRES REU

NSF-Funded Research Intern — Generative AI for Molecular Design

East Lansing, MI
May 2026 – Jul 2026

- Selected as 1 of 10 students from 200+ applicants for the NSF-funded ACRES REU to conduct machine learning research under Dr. Alex Dickson.
- Extended the LEVI molecular generation framework in PyTorch by developing SE(3)-equivariant graph neural network and diffusion-model methods for generating chemically valid drug candidates inside protein binding pockets.
- Improved distributed multi-GPU training and inference on Michigan State's high-performance computing cluster while evaluating new diffusion-model architectures for molecular inpainting and protein-conditioned generation.
- Developed **ACForce**, a novel graph-connectivity objective based on Laplacian graph theory that increased molecular connectivity by over **30%** and by nearly **50%** on benchmark drug-like molecule datasets while preserving molecular quality.
- Presented research at the MSU Mid-SURE Conference ([poster](#)), earned **1st Place Presentation** at the ACRES Symposium, produced rigorous theoretical analysis of the proposed method, and served as the **first-author of one paper** and the **second-author of another** currently being prepared for submission to NeurIPS and ICLR.

Brown Lab @ Vanderbilt University Department of Pharmacology

Protein Dynamics & Cheminformatics Research Intern

Nashville, TN
Jun 2023 – Jun 2025

- Built GNN & VAE workflows in PyTorch for computational drug discovery across nearly **2 million** drug-like molecules under Dr. Benjamin Brown.
- Developed protein dynamics simulations and transfer-learning models that achieved **85% predictive accuracy** for molecular metabolic stability prediction.
- Presented research findings in weekly lab meetings and collaborated on computational workflows spanning molecular simulations, machine learning, and cheminformatics.
- Named a **Regeneron STS 2025 Top 300 Scholar** for “[Contextualising Data-Sparse QSPR Property Prediction with Categorical Representations Built from Variational Autoencoders](#)”, with the work accepted to the Princeton Undergraduate Research Journal.

EXTRACURRICULAR INVOLVEMENT & PROJECTS

Princeton Puzzles Club

Officer & Princeton Puzzlehunt Puzzle Writer/Tester

Sep 2025 – Present

- Design and test competition-grade logic puzzles by constructing multi-step solution paths, verifying uniqueness, balancing solve difficulty, and identifying edge cases.
- Helped design and test three puzzles for Princeton Puzzlehunt 2026, supporting a competition with 500+ participants through live debugging, grading, and real-time hint management.

[Black-Scholes Options Pricing & Risk Analysis Platform \(Python\)](#)

Jun 2026

- Built a Python platform implementing the Black-Scholes model to price European options while analytically computing all five option Greeks.
- Validated analytical prices using Monte Carlo simulation across 100,000 geometric Brownian motion paths.
- Implemented Newton-Raphson optimisation to compute implied volatility directly from market prices.
- Generated live volatility surfaces from real SPY options-chain data, demonstrating the empirical volatility smile absent from classical Black-Scholes assumptions.
- Visualised option sensitivities, multi-leg trading strategy payoffs, and mark-to-market portfolio P&L across simulated market scenarios.

[GridSmart: Real-Time Grid Optimisation & Smart Home Scheduling](#)

Apr 2026

Princeton University Claude Builder Club Hackathon

- Built an AI-powered electricity optimisation platform integrating ERCOT pricing, SmartThings, and Alexa APIs to automatically schedule household energy usage based on cost, renewable availability, and grid demand.
- Designed an LLM-powered data ingestion pipeline and interactive dashboard that optimised electricity costs while analysing regional energy equity and renewable utilisation.
- Presented the project to Princeton ORFE and Computer Science faculty during the final judging round.

SKILLS, AWARDS & DISTINCTIONS

Languages, Tools, and Proficiencies: Python, Java, R, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, L^AT_EX, Excel, PowerPoint

Certification: [J.P. Morgan Quantitative Research Job Simulation](#) (Forage)

Honours:

- National Merit \$2500 Scholarship Recipient (2025) | Regeneron STS Top 300 Scholar (2025)
- U.S. Presidential Scholar Semifinalist (2025) | Coca-Cola Scholarship Semifinalist (2025) | Hagan Scholarship Finalist (2025)
- Co-author, *PLOS Biology*: “[Developing Inhibitory Peptides Against SARS-CoV-2 Envelope Protein](#)”
- Quiz Bowl NAQT Nationals Qualifier (2024, 2025); Princeton Quiz Bowl A-Team (Top 10, ACF Winter Regionals)
- 2× IAC National Science Bee Nationals Qualifier (2024, 2025), national placements of 15th and 4th.